

KHOVALING, TajikistanWMO: **388460**Lat: **38.367N**Lon: **69.983E**Elev: **4816**StdP: **12.31**Time Zone: **5.00 (E05)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	14.4	5.4	8.7	9.9	10.2	11.1	15.9	16.4	29.8	15.9	27.9	10.6	30	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	93.1	73.8	90.6	72.1	88.3	70.9	76.1	89.0	74.2	87.8	72.5	86.3	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	144.6	85.2	70.2	133.2	83.9	68.1	123.9	83.2	43.7	88.8	41.7	88.5	39.9	86.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	14.5	13.5	3.7	96.8	8.0	2.1	-2.1	98.3	-6.8	99.5	-11.3	100.6	-17.1	102.1		
(5)			3.3	77.6	7.9	2.2	-2.5	79.2	-7.1	80.5	-11.5	81.8	-17.3	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	31.9	33.0	45.1	54.8	62.4	70.8	77.2	75.3	68.8	57.4	44.6	35.2
	DBStd	17.17	7.59	9.54	7.93	6.45	4.56	4.82	3.76	3.57	4.36	6.60	8.24	8.80
	HDD50	1926	561	479	185	30	0	0	0	0	0	19	192	460
	HDD65	4760	1026	896	617	309	105	9	0	0	16	246	612	924
	CDD50	3688	0	3	34	173	386	623	843	784	564	247	30	1
	CDD65	1047	0	0	0	2	26	183	378	319	130	9	0	0
	CDH74	11375	0	0	3	68	387	1982	4317	3305	1156	151	6	0
	CDH80	4444	0	0	0	5	65	698	1989	1361	305	21	0	0
Wind	WSAvg	7.6	7.7	7.6	7.5	7.3	7.4	7.5	7.4	7.7	8.0	7.9	7.6	7.7
Precipitation	PrecAvg	20.4	2.1	2.8	3.3	3.1	2.4	1.0	0.6	0.3	0.3	0.9	1.7	2.0
	PrecMax	29.0	3.8	4.8	5.3	7.4	4.7	2.1	1.5	0.8	0.7	2.0	4.4	3.9
	PrecMin	12.8	0.9	0.9	0.9	1.0	0.4	0.4	0.2	0.1	0.0	0.2	0.5	0.3
	PrecStd	4.0	0.8	0.9	1.1	1.2	1.1	0.4	0.3	0.2	0.2	0.4	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	66.8	71.0	80.1	85.6	92.6	96.9	94.1	87.5	82.8	72.7	61.6
		MCWB	45.3	51.8	57.4	64.2	66.9	71.1	74.5	75.3	67.3	64.3	56.5	48.6
	2%	DB	51.1	55.0	66.0	75.5	81.5	89.3	94.6	91.3	85.6	78.2	66.0	56.4
		MCWB	42.3	47.1	53.7	60.6	66.1	69.8	74.3	73.5	66.6	61.9	54.4	46.2
	5%	DB	46.3	49.5	61.5	71.9	78.2	86.9	92.0	89.6	83.7	74.0	61.3	51.9
		MCWB	40.8	43.4	51.9	59.2	64.6	70.1	73.1	71.8	65.9	59.8	50.5	43.6
	10%	DB	42.5	45.3	57.7	67.8	75.2	84.3	89.4	87.4	81.3	70.2	56.8	47.2
		MCWB	38.6	41.3	49.8	57.6	63.6	69.0	71.7	70.7	65.0	58.0	48.5	41.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	52.0	59.6	66.4	71.2	76.2	78.3	78.4	70.4	65.8	58.9	49.9
		MCDB	52.3	62.5	67.0	75.6	79.1	86.3	90.8	90.0	83.6	76.1	71.1	58.2
	2%	WB	43.9	47.3	55.4	63.7	68.0	73.2	76.2	76.3	68.8	63.6	54.2	46.9
		MCDB	47.9	53.3	63.8	71.2	76.6	84.7	89.9	88.4	82.4	75.2	64.0	54.6
	5%	WB	41.2	44.4	52.8	61.0	66.3	71.5	74.7	73.8	67.0	61.4	51.4	44.2
		MCDB	45.0	48.7	59.6	68.4	75.2	83.7	89.1	87.7	81.1	72.5	59.0	50.6
	10%	WB	39.0	41.9	50.5	58.5	64.6	69.8	73.2	71.8	65.3	58.8	49.2	41.7
		MCDB	42.5	44.9	56.1	66.1	73.3	81.4	87.6	85.8	79.7	68.6	56.0	46.6
Mean Daily Temperature Range	5% DB	MDBR	16.1	17.0	17.2	19.2	21.3	23.5	23.4	22.4	22.3	21.0	18.7	17.0
		MCDBR	20.1	21.3	23.9	26.8	27.1	26.9	25.3	25.1	25.4	27.9	26.1	22.4
		MCWBR	13.4	14.2	15.0	16.0	15.5	15.0	13.7	14.6	14.7	16.1	15.2	14.2
	5% WB	MCDBR	17.3	19.3	22.0	23.7	24.6	26.1	24.1	24.0	22.8	24.6	24.4	20.2
		MCWBR	13.0	13.9	15.6	16.6	16.6	17.6	15.4	15.9	14.7	16.1	15.8	14.0
Clear-Sky Solar Irradiance	taub		0.331	0.377	0.439	0.454	0.466	0.488	0.483	0.443	0.403	0.420	0.384	0.332
	taud		2.180	2.059	1.953	1.971	1.981	1.923	1.934	2.038	2.111	2.004	2.067	2.175
	Ebn at Noon		261	262	257	262	261	254	254	262	266	243	236	250
	Edh at Noon		34	44	54	56	57	60	59	52	45	46	38	32
All-Sky Solar Radiation	RadAvg		724	955	1328	1668	1998	2353	2420	2263	1893	1250	767	628
	RadStd		64	128	113	180	161	135	99	90	72	129	115	61